

AMS / MagEAM — Blueprint Completo del Sistema

Versión: 1.0 · 2026-05-12

Owner: David Cabezas (VSC)

Cliente principal: Cliente Minero CL-SN (anonimizado)

Stack: React 19 + Vite · FastAPI + SQLAlchemy 2.0 · SQLite · Docker · nginx

Deploy: mageam.com (Hetzner VPS) · HEAD cd83670

Estadísticas

- 57 routers FastAPI
- 86 services Python
- 67 modelos BD
- 562 endpoints API
- 4 agentes IA + 36 skills
- 1593 tests pytest

1. Arquitectura de alto nivel

1.1 Componentes principales

Capa	Componente	Detalle
Usuarios	Web App / Field PWA	7 roles (Operador, Mantenedor, Supervisor, Planner, Programador, Ing. Confiabilidad, Manager, Admin)
Edge	nginx 1.29-alpine	TLS termination + reverse proxy port 443
Frontend	React 19 + Vite	Build estático servido por nginx
Field PWA	React/Vite mobile	Offline-first para captura en patio
Backend	FastAPI 0.110 (Py3.13)	562 endpoints en 57 routers
Storage	SQLite (datafile) + LanceDB	67 tablas + RAG vectors
AI agents	4 agentes Anthropic	Orchestrator + Reliability + Planning + SpareParts
External	SAP PM (cliente, futuro)	Stub queue con worker

1.2 Flujo de request típico

1. Cliente HTTPS → nginx (443)
2. nginx route → `/api/v1/*` al backend FastAPI, `/` al frontend estático, `/field` al PWA
3. Backend: `Depends(get_db)` + `Depends(get_current_user)` (JWT validation)
4. Service layer ejecuta business logic + audit log

- 5. ORM ↔ SQLite (o LanceDB para búsquedas RAG)
- 6. Respuesta + WebSocket broadcast cross-tab

2. Backend (FastAPI)

2.1 Routers (57 archivos)

Routers principales agrupados por dominio:

Dominio	Routers
Capture & WR	`/capture`, `/work-requests`, `/media`
OT	`/managed-work-orders`, `/work-orders`
Scheduling	`/scheduling`, `/preparativos`, `/assignments`
Planning	`/planner`, `/backlog`, `/workflow`
Reliability	`/fmea`, `/criticality`, `/hierarchy`, `/rca`, `/reliability`
Reporting	`/reporting`, `/dashboard`, `/analytics`, `/analytics_dashboards`
System	`/admin`, `/auth`, `/mfa`, `/security`
AI	`/ai-agents`, `/agentic`, `/rag`, `/troubleshooting`, `/programmer_agent`, `/supervisor_agent`, `/planificador_agent`
Integration	`/sap-pm`, `/sap`, `/catalogs`, `/data-import`, `/dms`, `/sync`
Misc	`/improvement-actions`, `/execution`, `/post-maintenance`, `/feedback`, `/notifications`, `/contractors`, `/expert_knowledge`, `/capture_geo`, `/financial`, `/imports`, `/or_projects`, `/sales`, `/transcribe`, `/work_packages`, `/tasks`, `/reports_export`, `/execution_checklists`, `/sprint6_scaffolds`

2.2 Services (86 archivos)

Categoría	Servicios principales
Core	`managed_wo_service`, `work_request_service`, `capture_service`, `audit_service`
Scheduling	`scheduling_service`, `planner_service`, `backlog_service`
Reliability	`fmea_service`, `rca_service`, `reliability_service`, `criticality_service`
RAG / AI	`rag_service`, `troubleshooting_service`, `agentic_`* (12 archivos)
Execution	`execution_service`, `post_maintenance_service`
Integration	`sap_export_service`, `dms_service`, `data_import_service`
Auth	`auth_service`, `permissions_service`

2.3 Engines deterministas (tools/engines/)

Lógica computacional pesada sin LLM:

- `priority_engine` (R8 + GFSN scoring)

- `criticality_engine` (matrix risk class)
- `rcm_decision_tree` , `fmea_engine`
- `weibull_engine` , `pareto_engine` , `jackknife_engine`
- `planner_engine` , `backlog_optimizer`
- `shutdown_engine` , `resource_leveling`
- `ganttt_generator` , `scheduling_engine`
- `kpi_engine` , `de_kpi_engine` , `planning_kpi_engine`
- `quality_score_engine` , `capa_engine`
- `variance_detector` , `trace_engine` , `feedback_engine`
- `deliverable_writer` , `rbi_engine` , `moc_engine`
- `health_score` , `lcc_engine` , `material_mapper`
- `equipment_resolver` , `equipment_library`

3. Frontend (React 19 + Vite)

3.1 Páginas principales (38 archivos)

Categoría	Páginas
Dashboards	Dashboard, ExecutiveDashboard, AnalyticsPage, PerformanceAnalysis
Captura	FailureCapture, FailuresEvents, FieldCapture (PWA), EquipmentChat
OT	WorkOrdersPage, WorkManagement, Planning, Execution, ExecutionChecklists, PostMaintenance
Scheduling	Scheduling (v2 Horarios), SupervisorBoard, Backlog
Reliability	FMEA, FMECA, Criticality, Reliability, RCA, RagLab, DefectElimination
System	Settings, SettingsPage, Profile, Team, AuditLogPage, Admin, DataImport, DocumentManagement
AI	AIAgents, AgenticCapabilities, ProgrammerAgent, SupervisorAgent, PlanificadorAgent
Misc	ImprovementActions, ContractorsPage, ProjectSelector, ExpertKnowledge, SapPmPage
Mobile	mobile/MobileCreateWR

3.2 Componentes compartidos

`Header` , `Sidebar` , `WelcomeCard` (tutorial v2 7 steps), `Toast` , `ConfirmDialog` , `SmartCaptureModal` , `SmartAssignModal` , `CancelWOModal` , `ExecutiveView` , `TacticalOperationsView` , `PageGuard` (RBAC), `WsStatusIndicator` , `PriorityBadge` , `KPICard` , `HelpPopover` , `FeedbackWidget` , `LoadingSpinner` .

3.3 Hooks + Contexts

- `useAuth` (AuthContext) — JWT + refresh
- `useWebSocket` — auto-reconnect, broadcast subscribe
- `usePermissions` — RBAC scoping (6 roles × 24 modules)
- `useLanguage` (LanguageContext) — i18n EN/ES/FR/AR
- `useToast` , `useConfirm`

3.4 API client (`api.js`)

Wrapper sobre `fetch` con:

- Inyección automática `Authorization: Bearer <jwt>` desde `localStorage`
- Refresh token transparente al recibir 401
- WS `client_id` en headers (cross-tab session tracking)
- Errors → throw con `r.statusText` + body

4. Sistema de 4 Agentes IA

4.1 Configuración

ID	Agente	Modelo	Max turns	Milestones
AG-001	Orchestrator	Sonnet 4.5	20	All
AG-002	Reliability Engineer	Opus 4.6	40	M1, M2, M3
AG-003	Planning Specialist	Sonnet 4.5	30	M3, M4
AG-004	Spare Parts Specialist	Haiku 4.5	15	M3

4.2 Skills por agente

Agente	Skills principales
AG-002 Reliability	build-hierarchy, classify-criticality, fmeca-build, rcm-decide, pareto-jackknife, weibull, moc-manage, rbi-assessment
AG-003 Planning	assemble-work-packages, group-backlog, calculate-priority, schedule-weekly-program, orchestrate-shutdown, export-to-sap, analyze-work-order (SF-661 v0.2 nuevo), manage-capx, calculate-planning-kpis, calculate-life-cycle-cost, optimize-cost-risk, generate-execution-checklists
AG-004 Spare Parts	suggest-materials, optimize-spare-parts-inventory
AG-001 Orchestrator	gestiona los milestones M1-M4 y coordina los specialists

4.3 Los 4 Milestones

Milestone	Nombre	Agentes	Deliverables clave
M1	Hierarchy + Criticality	AG-002	Equipment hierarchy tree, criticality matrix

M2	FMECA + RCM Decisions	AG-002	FMECA table, RCM decision sheets
M3	Tasks + Work Packages + Materials	AG-003 + AG-004	Task list, work packages, work instructions, material assignments
M4	SAP Export Package	AG-003 + AG-001	SAP upload package (DRAFT), validation report, strategy summary

Cada milestone tiene **human approval gate** explícito.

5. Flujo de vida end-to-end de una OT

Paso	Actor	Acción	Endpoint / Vista	Resultado
1	Técnico (móvil)	Foto + audio + GPS en patio	Field App PWA	FieldCapture (capture_id)
2	Sistema	POST /capture/	capture_service	WorkRequest (DRAFT, AV-NNNNN)
3	Supervisor	Aprueba / rechaza WR	PUT /work-requests/{id}/validate	WR.status = APROBADO + auto-add backlog
4	Planner	Crea OT desde WR	POST /managed-work-orders/from-wr	ManagedWorkOrder (CREADO, OT-2026-NNNNN)
5	Planner	Define operaciones + materiales	PUT /managed-work-orders/{id}	operations[], materials[] poblados
6	Planner	Avanza CREADO → PLANIFICADO	PUT /release + /plan	status=PLANIFICADO
7	Planner	Marca para programar	PUT /managed-work-orders/{id} {status:EN_PROGRAMACION}	status=EN_PROGRAMACION
8	Programador	Drag-drop al calendario	PUT /scheduling /managed-work-orders/{id}/schedule	status=PROGRAMADO + planned_start/end + assigned_workers
9	Programador	Reservar semana	POST /scheduling/reserve-week	Bloqueado si overcap
10	Técnico	Inicia OT	PUT /start	status=EN_EJECUCION + actual_start
11	Técnico	Notif HH parcial	PUT /progress	operations[i].actual_hours acumula
12	Técnico	Completa ops	PUT con operations actualizadas	completion_pct=100
13	Supervisor	Cierra con firma	PUT /close + signature + pin + gate_acks	status=CERRADO + closed_by_pin_hash
14	Sistema	Audit log final	audit_service.log_action(CLOSE)	audit_log entry
15	Engineer (opcional)	RCA + lessons learned	POST /rca/draft-from-wo	RCA + RAG ingest
16	Engineer	Improvement actions	POST /improvement-actions	CAPA generadas

5.1 Pre-close gates (SF-570)

Gate	Tipo	Bloquea cierre?
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ALL_OPS_DONE	auto	sí
OPS_HH_NOTIFIED	auto	sí
HH_VARIANCE_OK ($\leq 25\%$)	auto	sí (override permitido con razón)
MATERIALS_OK	auto	no
SUPERVISOR_QA	manual	sí (supervisor debe ackear)
NO_OPEN_NOTIFS	auto	no

6. RAG (Retrieval Augmented Generation)

Componente	Detalle
Backend	LanceDB embedded en <code>`data/lancedb/`</code>
Embed model	<code>`paraphrase-multilingual-MiniLM-L12-v2`</code> (sentence-transformers, 384 dims)
RAM idle	0 MB (lazy load)
RAM al primer query	~1.5 GB (modelo cargado en memoria)
Tablas	<code>`ot_history`</code> (56 OTs cerradas), <code>`lessons_learned`</code> , <code>`manuals`</code>
Chunking	word-based, 500 tokens max, 50 tokens overlap
API	<code>`add_chunks`</code> , <code>`search(table, query, k=5)`</code> , <code>`embed`</code> , <code>`stats`</code>
Service	<code>`api/services/rag_service.py`</code>
Casos de uso	RCM-Strategy (#33), Shift-Handover (#34), Post-Maint-Learn (#35), KB-Curator (#40)

7. Permission Matrix (RBAC)

7.1 Roles (6)

Rol	View Tier	Edit Scope
admin	admin	all
manager	executive	executive
planner	tactical	planning
engineer	tactical	reliability
supervisor	tactical	execution
tecnico	tactical	execution

7.2 Implementación

- **Frontend:** `usePermissions.js` con 24 módulos × 2 acciones (view, edit)
- **Backend:** `require_role(*roles)` decorator en routers
- **IDOR-scoping:** si `user.role NOT IN (admin, manager)` → filter por `user.plant_id` en query

7.3 Módulos públicos vs admin-only

Módulo	Roles que ven
dashboard	todos
work-requests	todos (view), admin/planner (edit)
scheduling	admin/planner/engineer/tecnico/supervisor
audit-log	admin/manager (full), supervisor (su planta), planner (propio)
agentic-capabilities	**solo admin**
ai-agents	admin/manager/engineer
data-import	admin/manager
settings	todos

8. Auditoría + Inmutabilidad (SF-660)

8.1 Cobertura

14 entidades cubiertas:

work_request, managed_work_order, operations, materials, support_equipment, user, settings, backlog, weekly_program, criticality_assessment, fmeca_worksheet, rca, dms_documents, plus agent actions.

8.2 Política

Item	Valor
Retención activa	2 años
Retención archivo	5 años (export CSV semanal)
Eliminación	>5 años con notificación previa
Inmutabilidad	append-only · tests pytest validan no hay DELETE en código
Script purga	`scripts/purge_audit_log_5y.py --dry-run` por default

8.3 Acceso por rol

Rol	Acceso
admin / manager	todo
supervisor	su planta (vía payload.plant_id) + propias acciones
planner / engineer / tecnico	solo propias acciones
otros	403

9. Despliegue (VPS Hetzner — mageam.com)

9.1 Infraestructura

Item	Detalle
Hosting	Hetzner VPS
IP	187.77.223.137
Dominio	mageam.com
Stack	Docker Compose
TLS	Let's Encrypt cert (renovación auto)

9.2 Containers

Container	Imagen	Puerto	Función
ocp-nginx	nginx:1.29-alpine	80, 443	TLS termination + reverse proxy
ocp-backend	python:3.13 (FastAPI)	8000	API 562 endpoints
ocp-frontend	nginx (Vite build)	5173	Static React build

9.3 Volumes

Volumen	Mount	Contenido
ocp_db_data	/app/data	SQLite + audit + capture photos
ocp_sap_data	/app/sap_mock/data	SAP mock CSV
ocp_or_deliverables	/app/or_deliverables	OR-System deliverables (legacy)
ocp_hf_cache	/home/appuser/.cache/huggingface	Embedding model cache (~1.5 GB)
ocp_dms_docs	/usr/share/nginx/dms	DMS documentos servidos por nginx

10. CI/CD pipeline

10.1 Github Actions (`.github/workflows/ci.yml`)

Job	Steps	Bloquea merge?
backend	pytest smoke critical	sí
backend	pytest full API suite	no (warnings ok)
frontend	npm install + build + lint	sí (build)
docker	Build backend image	sí

10.2 Deploy flow

```
```bash
```

## Local

```
git push origin main
```

```
git push vps main
```



## VPS (auto o manual)

```
ssh root@mageam.com
```

```
cd /root/ASSET-MANAGEMENT-SOFTWARE
```

```
docker compose build ocp-backend ocp-frontend
```

```
docker compose up -d ocp-backend ocp-frontend
```

```
docker compose restart ocp-nginx
```

```
...
```

Tiempo total deploy: 15-20 segundos.

## 11. Decisiones de diseño claves

Decisión	Por qué
**JSON embebido en `operations/materials/workers`**	Snapshot inmutable + flexibilidad schema cliente
**Sin FK formal en `work_request_id`**	OTs fast-track P1 sin WR previo + robustez a borrados
**Audit log universal** (entity_type + id)	1 tabla vs N tablas, query simple cross-entity
**4 agentes específicos vs 1 mega-agente**	Especialización + token budget separado por rol
**LanceDB embedded vs vector DB externo**	0 ops overhead, RAG funciona sin servicios extra
**SQLite vs Postgres**	Single-tenant inicial, deploy simple; migrar fácil cuando escale
**Optimistic locking (version + If-Match)**	UX sin lock contention vs pessimistic
**Stub queue para SAP**	Cliente sin credenciales aún; pipeline listo
**Lazy load embed model**	Idle 0 RAM; 1.5GB solo al primer query
**Scoped roles via JWT plant_id**	IDOR-mitigation desde JWT, no query param

## 12. Hitos completados (commits recientes)

### Sprint 7 VSC (en curso)

Ticket / Tema	Highlights
SF-654	Eventos sistema en tab Comentarios
SF-655	Foto + audio en OT comments (Web Speech API + MediaRecorder)
SF-656	Auditoría visual capacidad + violaciones turno DAY/NIGHT
SF-657	Endpoint `orphans` (6 tipos huérfanas)
SF-659	Política OT-Calendar sync documentada
SF-660	Audit log policy + tests inmutabilidad + script purga 5y
SF-661 v0.2	IA Analyze OT — 4/7 funciones deterministas
SF-662	Preparativos OT Rappi (modelo + 4 endpoints + 13 tests)

0A2	Gap Analysis Magdalena (174 items, 95.1% cobertura)
0D1	Anonimización completa Goldfields (plant alias local)
0G2	Supervisor role + Agentic admin-only
Tests	12 → 0 fails en `test_api` (212/214 passing)
Tutorial	Welcome card v2 7 steps
Scheduling v2	Vista Horarios eje vertical + matching auto por skill

## 13. Roadmap SP8

- SF-661 funciones 2/4/7 — req histórico real (0B2)
- SF-662 UI completa — req spec final Jorge
- 0E centros de costo + clases de gasto
- 0G1 casos de uso Carlos — input para E2E Playwright
- Refactor módulo Ejecución (deadline VIE 15-may)
- vitest + Playwright CI nightly
- S3/blob storage para fotos
- SAP integration real cuando lleguen credenciales

## 14. Referencias cruzadas

Doc	Path
Modelo relacional OT	[docs/MODEL_RELATIONSHIPS.pdf](MODEL_RELATIONSHIPS.pdf)
Database blueprint completo	[docs/DATABASE_BLUEPRINT.pdf](DATABASE_BLUEPRINT.pdf)
Política audit log	[docs/AUDIT_LOG_POLICY.pdf](AUDIT_LOG_POLICY.pdf)
OT-Calendar sync	[docs/OT_CALENDAR_SYNC.pdf](OT_CALENDAR_SYNC.pdf)
Roadmap SP7/SP8	[docs/TANDAS_SP7_SP8_2026-05-11.pdf](TANDAS_SP7_SP8_2026-05-11.pdf)
Gap analysis 174 items	[1-output/M3/gap-analysis-magdalena-2026-05-12.xlsx](../1-output/M3/gap-analysis-magdalena-2026-05-12.xlsx)
Testing strategy	[1-output/M3/testing-strategy-magam-2026-05-12.pdf](../1-output/M3/testing-strategy-magam-2026-05-12.pdf)
BBP SAP Anglo	[docs/AMSA_BBP_PM_04_Rev_0.docx](AMSA_BBP_PM_04_Rev_0.docx)
Blueprint agentes	[skills/00-knowledge-base/architecture/BUEPRINT.md](../skills/00-knowledge-base/architecture/BUEPRINT.md)

Commit: `cd83670`

